Quantifying information gain in narrative stimuli using open-source large language models

Aidan Goeschel[1,2*] **Rohin A. Palsule**[2*] **Jungsun Yoo**[3] **Janice Chen**[4] **Aaron M. Bornstein**[2,5,6] **Ari Khoudary**[2,5,6+]

[1] Department of Computer Science, University of California, Irvine
[2] Department of Cognitive Sciences, University of California, Irvine
[3] Neuroscience Institute, Princeton University
[4] Department of Psychological & Brain Sciences, Johns Hopkins University
[5] Center for the Neurobiology of Learning and Memory, University of California, Irvine
[6] Center for Theoretical Behavioral Sciences, University of California, Irvine
[*] These authors contributed equally.

agoesche@uci.edu, rapalsul@uci.edu, janice@jhu.edu, aaron.bornstein@uci.edu, ari.khoudary@uci.edu

Abstract

Narrative structure is central to human memory, scaffolding the consolidation of past experience into structured knowledge and shaping the encoding of to-be-remembered information. Further, narratively-structured stimuli increase neural signals of engagement and value relative to non-narrative stimuli, suggesting that humans may exhibit an inductive bias toward narratively-structured information. These two lines of research have recently been integrated into a theory proposing that narrative structure facilitates *credit assignment* in high-dimensional and temporally discontiguous state-spaces (Chen & Bornstein, 2024). Because credit assignment is a ubiquitous problem faced by cognitive agents, understanding how it is solved by humans can help advance understanding of how it can be solved by other agents. To do this, methods are needed for quantifying the time-evolving valuation of information as it unfolds within a narrative. A recently-developed computational metric called *Sequentiality* (Sap et al., 2022) appears to be a promising tool toward this end, but additional work is needed to assess its utility for applications to the question of credit assignment. This project addresses that gap by (1) creating and validating an open-source, ensemble-based implementation of *Sequentiality*, (2) systematically characterizing the role of prior knowledge in shaping *Sequentiality* values, and (3) establishing a direct empirical link between *Sequentiality* and human judgments of event importance and causal centrality. Expanding upon the findings of Sap et al. (2022), we show that *Sequentiality* values critically depend on the prior knowledge an observer has about narrative structure. We suggest these findings motivate interpreting the metric in terms of *contextual information gain*, rather than the original interpretation of schema-consistency or familiarity. Together, these findings pave the way for using Sequentiality as a proxy for a key theoretical construct involved in solving the credit assignment problem: an eligibility trace indicating which previous states are most relevant for updating or retrieving based on incoming information from the environment.

Introduction

All learning systems—humans, animals, and neural networks—are tasked with solving the *credit assignment* problem: determining which previous actions or events caused a current outcome. Accurately attributing credit to past events is a challenging computational problem whose complexity scales nonlinearly with the space of possible actions and the time elapsed between action and outcome. Because human learners are remarkably successful at appropriately assigning credit in complex, high-dimensional environments, identifying the basic algorithms and representations involved can shed important light on limitations in state-of-the-art methods currently used by artificial agents when dealing with credit assignment. Further, elucidating the computations underlying human credit assignment contributes to basic understanding of human cognitive psychology and neuroscience, which in turn can offer insight into why *errors* of human credit assignment take the forms they do.

Recently, Chen and Bornstein (2024) proposed that *narrative schema* stored in long-term memory might be an essential component of humans' ability to efficiently solve complex credit assignment problems. The authors suggest that narrative schema equip humans with complex prior knowledge about (1) the plausibility of event sequences across different contexts and (2) causal relationships among those events over different timescales. These narrative schema—or simply, *narratives*—are theorized to help humans reduce uncertainty about the (nearly infinite) space of possible solutions to complex credit assignment problems in two ways. First, they allow agents to *ignore* aspects of the environment that are irrelevant with respect to the space of tasks they are likely to encounter, simplifying the informational demands on long-term memory. Second, narratives help agents select *counterfactual simulations* that are most relevant to the present outcome in order to better reason about its plausible causes. This counterfactual selection function is especially important for increasing the tractability of credit assignment computations, and usage of counterfactual information has recently been shown to improve artificial agents' credit assignment abilities (Van Hasselt et al., 2021).

Chen and Bornstein (2024)'s theory was developed to account for mounting evidence from naturalistic neuroscience

In *Proceedings of the 9th Conference on Cognitive Computational Neuroscience*, New York, NY, USA, 2026. Copyright 2026 by the author(s). Licensed under CC BY 4.0.

research demonstrating that human memory systems are uniquely sensitive to causal information (Antony et al., 2024; Cohn-Sheehy et al., 2021; Lee & Chen, 2022; Lee et al., 2020; Zadbood et al., 2022). Crucially, these findings were driven by the use of *narrative stimuli*: auditory stories, animated videos, and television shows/movies. These stimuli are increasingly prevalent in naturalistic cognitive neuroscience because they approximate the complexity, contents, and statistics of real-world experience while also allowing a reasonable degree of experimental control and manipulation. A puzzling yet persistent finding in this domain is the activation of neural regions traditionally associated with *value* during the viewing of narrative stimuli (e.g., ventromedial prefrontal cortex; Baldassano et al., 2018; Chang et al., 2021; Zadbood et al., 2022). Chen and Bornstein (2024) suggest that these activations contain signals about what information to encode vs. ignore when processing complex, multimodal information in real time, as subjects are tasked with doing while watching narrative stimuli in the fMRI scanner.

The testing of this hypothesis has been held back by a lack of methods for quantifying the relative value of information while processing narrative stimuli in real-time. We aim to address this gap by building upon a recently-developed metric *Sequentiality* to quantify the "flow" of events in text-based narrative stimuli (Sap et al., 2020, 2022). The *Sequentiality* metric derives likelihood values for each event (or sentence) in a narrative using token probabilities obtained via large language models (LLMs). We believe this is a promising metric for quantifying event-level fluctuations in the informativeness of particular observations for a number of reasons. First, the massive scale of data on which LLMs are trained makes them reasonable proxies for ground-truth frequencies about token co-occurrences. Second, the attention mechanism in transformers is uniquely capable at identifying token relations across large windows of textual "evidence". Taken together, these properties of contemporary LLMs make them suitable candidates for investigating how complex, high-dimensional prior knowledge might shape real-time encoding of continuously unfolding information.

Original Sequentiality metric: formalization

Sap et al. (2022) introduced the *Sequentiality* metric to quantify the time-evolving log-likelihood ratio of consecutive sentences in a piece of written text. Specifically, the likelihood of a particular sentence s_i is defined by two generating models: one based on a user-defined topic τ only, and one based jointly on the topic and preceding h sentences. The original *Sequentiality* metric thus defined the "flow" c of a narrative text as the *difference* in the likelihood of sentence s_i between these two generating models:

$$c(s_i, h) = -\frac{1}{|s_i|}(logp(s_i|\tau) - logp(s_i|\tau, h)) \quad (1)$$

where s_i is sentence i, τ is the topic that sets the situation model for the story, and h is the history length defining number of previous sentences $s_{i-h:i-1}$ that are used to establish a local context; h is a free parameter ($h \in \mathbb{Z}^+$).

The first term inside the parentheses in Equation 1 corresponds to the likelihood of sentence s_i conditioned only on topic τ, and the second term corresponds to the likelihood of sentence s_i conditioned jointly on topic τ and previous context h. Essentially, this measure quantifies how much the previous sentences in the context *reduced* the surprise of observing sentence s_i. A positive *Sequentiality* value for a sentence s_i thus indicates that the previous h sentences made the probability of sentence s_i *greater* than expected by the topic alone, whereas negative values indicate that the previous h sentences made the probability of s_i *lower* than what was expected based on the topic (Sap et al., 2022). Finally, the metric is normalized by sentence s_i to control for differences in sentence lengths within the narrative.

Empirical support: original *Sequentiality* metric

The validity of *Sequentiality* was assessed by applying it to the HIPPOCORPUS dataset, which consists of short, diary-like stories about memorable life experiences (Sap et al., 2020). Participants were asked to recall an autobiographical event that happened to them recently, retell this story 3-6 months later, and/or narrate an imagined short story about topics derived from other participants' autobiographical recall. Critically, the authors used participants' self-generated event summaries as the topic τ for each narrative. To generate these topics, participants were instructed to provide a "short phrase describing the main event described". Sap et al. (2022) report two key findings as validation for *Sequentiality* capturing the "flow" of imagined and autobiographical stories (Figure 1A). First, imagined stories have the highest *Sequentiality* values relative to retold and recalled. Second, the *Sequentiality* of an autobiographical story *increases* upon being re-told several months later. The authors interpret these findings in the framework of schema-driven recollection and imagination, such that greater *Sequentiality* values indicate stories that unfold in a more schema-consistent manner over time (Sap et al., 2022).

Limitations of original findings

For our desired application of testing a *value of information* hypothesis about the dynamic encoding of narrative stimuli, the original findings are limited both methodologically and theoretically. The first methodological limitation is that the likelihoods used to compute the original *Sequentiality* findings (GPT-3) are no longer publicly available. The second limitation is that the original metric relies *only* on likelihoods obtained from GPT-3. Although the authors report that "our findings with *Sequentiality* remain similar when using language models trained on other datasets" (Sap et al., 2022, p. 6), they do not provide quantitative data confirming this, nor do they take steps to mitigate potential idiosyncrasies stemming from the model's training set or architecture.

Theoretically, the original findings are also limited in two ways. First, the authors do not report any systematic investigation into the role that *topic* plays in shaping *Sequentiality* values. Because our goal is to use the metric to test how complex, high-dimensional prior knowledge directs the apportioning of value signals during encoding, it is important to understand how the "prior" in the metric (i.e., the topic) affects the likelihoods of sentences across different narrative generating sources. Doing so would allow us to create a metric that captures, at some degree of resolution, the *difference* in valuation signals between an observer who has "full" vs. partial knowledge of narrative structure outcome prior to encoding the narrative into memory. This distinction is especially important because it mimics manipulations frequently used in the naturalistic neuroscience literature (Lee & Chen, 2022).

The second theoretical limitation pertains to how *Sequentiality* was interpreted with respect to the cognitive constructs it may help explain. The original authors interpret greater values of *Sequentiality* for imagined stories as reflecting an increased "flow" or coherence of information over time. This interpretation implies that *Sequentiality's* relationship to event importance ratings and subsequent memory can be explained in terms of the cognitive construct of *fluency* (Dantlgraber et al., 2019; Rajaram & Geraci, 2000), or ease of processing due to stimulus familiarity. However, this interpretation is based only on *Sequentiality* values defined *relative* to user-specified topics for each entry in HIPPOCORPUS. Alternative accounts of event importance and memorability suggest that *prediction error*—or, equivalently, information gain—should be the driving construct (DuBrow et al., 2017; Rouhani et al., 2020). These competing theoretical interpretations yield a testable prediction: if *Sequentiality* is capturing fluency, then schema-consistent information should always result in *higher Sequentiality* values. However, if *Sequentiality* is capturing prediction error, then schema-*in*consistent information should yield higher *Sequentiality* values. We evaluate these hypotheses using narratives from the HIPPOCORPUS and FILMFEST datasets, as well as a bespoke narrative from Chen and Bornstein (2024) designed to demonstrate the effect of prior knowledge on credit assignment for outcomes of complex, long-timescale events.

Methods: Open-source Implementation

To address the methodological limitations, we first develop and validate an open-source version of the *Sequentiality* metric. Importantly, our implementation automatically generates likelihoods from five different LLMs with varying architectures, training sets, and fine-tuning protocols, allowing for more robust conditional probability estimation and/or investigation into systematic variations in *Sequentiality* as a function of model provider. Our implementation of *Sequentiality* obtains sentence likelihoods from five different LLMs made available via the HuggingFace API. To address potential problems with bias in LLM training data and to mitigate the effect of differences in model architecture between models, we propose using an ensemble of LLMs. Specifically, our implementation utilizes likelihoods from SAKANA TINYSWALLOW 1.5B, GPT2-XL, OLMO-2-13B, LLAMA-3.1-8B, and QWEN2.5-7B. The Python code used to implement this open-source *Sequentiality* metric is available for public use on GitHub.

To calculate *Sequentiality*, we first tokenize the entire story using the same alphabet as each LLM in the ensemble. To obtain topic-only sentence likelihoods, $p(s_i|\tau)$, we append a preamble to our input text that tells the LLM to adjust its probabilities according to a user-defined topic τ. Then, for each token t in sentence s_i, we pass in all tokens before t in s_i and extract the distribution for the predicted next token. We can search through this distribution to find the log-probability of token t given the previously inputted tokens. These log-probabilities are summed to generate the topic-driven likelihood, $p(s_i|\tau)$.

To calculate the context-driven sentence likelihood, $p(s_i|\tau, h)$, we use the same preamble to establish a topic and the same mechanism of auto-regressive distribution generation. However, rather than inputting only s_i, we input the previous h sentences as well, where h is a hyperparameter that controls the history length of the model. The *Sequentiality* value for a single sentence $c(s_i, h)$ is then computed as specified in Equation 1: we compute the difference between topic and context likelihoods and then normalize by the number of tokens in the sentence.

Our method thus generates $h * M$ *Sequentiality* values for each sentence in a piece of written text, where h is the history length hyperparameter and M is the number of models in the ensemble. A primary strength of this method is the ability to quantify and represent uncertainty about *Sequentiality* values across model providers for a single sentence—or narrative—which was impossible to achieve with the previous approach. Further, as we show in Figure 1B, averaging across five open-source LLMs generates*Sequentiality* values that are nearly identical to those obtained from the now-proprietary GPT-3. This is notable for two reasons. First, the open-source nature allows for broader and more transparent application of the *Sequentiality* metric. Second, we make use of models that are orders of magnitude smaller than what was originally used by the authors, making the computation of *Sequentiality* more tractable for researchers without access to massive computational resources.

Implementation Validation

To confirm that our implementation accurately computes *Sequentiality*, we used the HIPPOCORPUS database to replicate the main findings of Sap et al. (2022). HIPPOCORPUS consists of 6,854 short, written stories about memorable life experiences. Human subjects were asked to recall an autobiographical event that happened to them recently, retell this story 3-6 months later, or create an imagined short story. For each story, participants also generated a short phrase describing what happens in the story. More details about the dataset can be found in Sap et al. (2020), and the dataset itself can be found on HuggingFace.

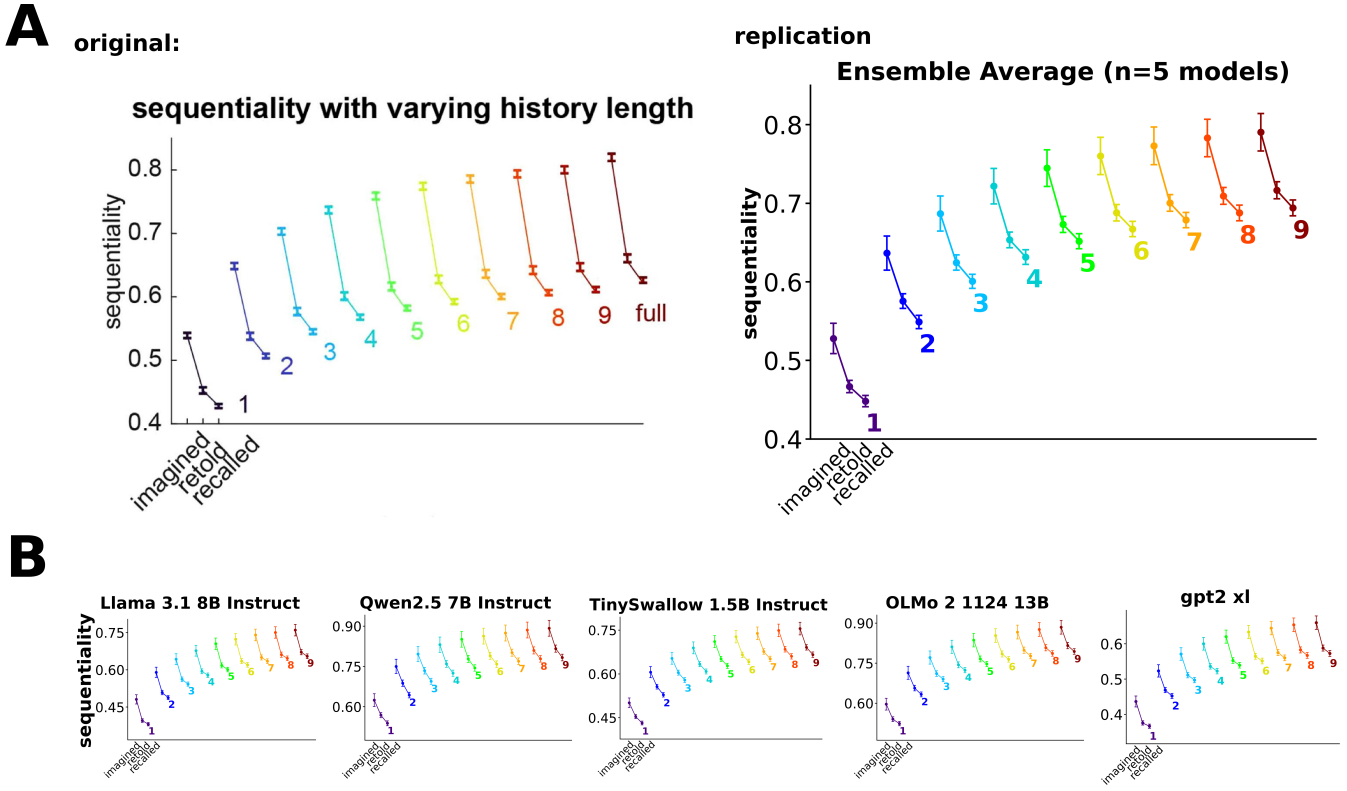


Figure 1: Main *Sequentiality* effect replication. (A) Mean *Sequentiality* values for stories in each condition as a function of history length. Error bars on the left plot (original) reflect standard error across stories, whereas error bars on the right plot (our implementation) reflect standard error across models. (B) *Sequentiality* values computed by each model in our ensemble.

We computed our ensemble *Sequentiality* metric for each sentence in each story in the HIPPOCORPUS dataset at history lengths h ranging from 1 to the full length of the story. Within each history length, we computed the total *Sequentiality* of a story as the mean of each sentence's *Sequentiality* as generated by each model. The total *Sequentiality* value at each level of history length is then computed as the mean across model-specific story-level *Sequentiality* values. Following the approach of Sap et al. (2022), we display and compare *Sequentiality* values only for topics that contain stories generated in each of the three conditions (recalled, retold, imagined; n=984 total stories). This approach controls for variability in *Sequentiality* due to topic, thus allowing for more direct comparisons of the effect of story generation type on *Sequentiality*.

As shown in Figure 1A, our implementation (right) almost perfectly replicates the original findings (left). A linear regression on *Sequentiality* values confirmed that we replicated the main effect of story type on *Sequentiality*. Both recalled ($\beta = -0.091$, $p < 0.001$) and retold ($\beta = -0.069$, $p = 0.001$) stories had significantly lower *Sequentiality* values than imagined stories. Additionally, history length monotonically increased *Sequentiality* across all story types ($\beta = 0.027$, $p < 0.001$). Further, Figure 1B shows that each model in our ensemble independently reproduced the original *Sequentiality* findings qualitatively, while exhibiting a moderate degree of variability in the precise *Sequentiality* values computed (note the different y-axes across figures). Importantly, Figure 1A shows that averaging across these individual estimates puts our ensemble metric back onto the same quantitative scale as the original *Sequentiality* values.

Metric Validation: Sensitivity to Event Sequences

Having confirmed that our implementation accurately computes *Sequentiality*, we next wanted to confirm that *Sequentiality* is indeed sensitive to the order of sentences—or events—in a narrative. Demonstrating sensitivity to event sequences is essential for validating the metric as a tool for automating credit assignment, especially since language models can be relatively insensitive to word order permutations (Chen et al., 2024).

Our analysis used the subset of HIPPOCORPUS stories with the topic "my wedding" (n=1 story per condition). We shuffled each story 100 times and then computed our ensemble *Sequentiality* metric for the shuffled stories at history lengths 1 and 30, with the latter being big enough to fit the entire story into the context. A linear model confirmed the re-

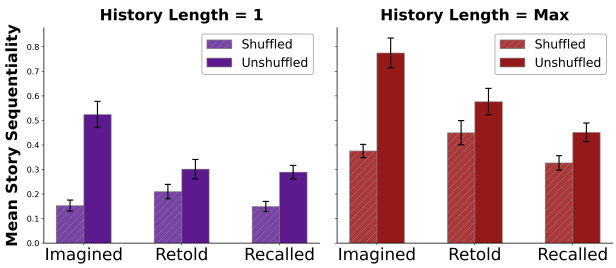


Figure 2: Sequentiality is sensitive to event sequences. Shuffling the order of sentences in HIPPOCORPUS entries for the topic "My wedding" removes effects of story type on mean *Sequentiality*. Error bars reflect standard error across models.

sults displayed in Figure 2: shuffling HIPPOCORPUS stories significantly decreased *Sequentiality* values computed with both minimum and maximum history lengths ($\beta = -0.3855$, $t_5 = -12.468$, $p < .001$). This result demonstrates that the mean *Sequentiality* value of a narrative is significantly impacted by the ordering of sentences within the narrative, empirically validating the use of *Sequentiality* to quantify the dynamics of narratively-structured information.

Theory: *Sequentiality* as a metric of contextual information gain

Now that the methodological limitations of past work have been addressed, we turn to addressing the theoretical question raised in the Introduction: can *Sequentiality* capture *information gain*, rather than familiarity? To answer this question, we use a combination of equivalence demonstrations, targeted case studies, and direct applications to narrative stimuli and human behavior.

Model Reformulation

This section demonstrates that the original *Sequentiality* formula Equation 1 can be equivalently re-expressed such that $c(s_i, h)$ corresponds to a metric interpretable as contextual information gain. We begin by combining the log terms in Equation Equation 1 to obtain the following ratio:

$$c(s_i, h) = -\frac{1}{|s_i|} log \frac{p(s_i|\tau)}{p(s_i|\tau, h)} \tag{2}$$

To better conceptualize the meaning behind this ratio, we can expand the terms in this equation using Bayes' Rule, which leaves:

$$c(s_i, h) = -\frac{1}{|s_i|} log \left(\frac{\frac{p(\tau|s_i)p(s_i)}{p(\tau)}}{\frac{p(\tau,h|s_i)p(s_i)}{p(\tau,h)}} \right) \tag{3}$$

Which simplifies to:

$$c(s_i, h) = -\frac{1}{|s_i|} log \left(\frac{p(h|\tau)}{p(h|\tau, s_i)} \right) \tag{4}$$

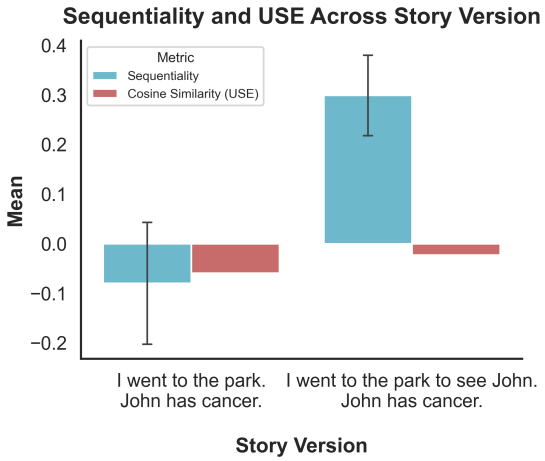


Figure 3: Minimal example demonstrating the notion of contextual information gain. *Sequentiality* values are greater for stories where the context contains information that makes the contents of sentence s_i relevant. When sentence s_i appears to have no relevance to the evolving context of a story, *Sequentiality* values drop. Error bars are standard error across models.

This final simplification represents how much information sentence s_i tells us about the context established in history h that is not already contained in topic τ, or in other words, the contextual information gain.

Case Study 1: Conceptualizing contextual information gain

Our first example was constructed to illustrate the notion of contextual information gain. We engineered two two-sentence stories that ended with exactly the same sentence, but differed by three words in the first sentence (see the x-axis in Figure 3 for the full stories). The key manipulation was whether the first sentence of the story loaded the character John into the context before *Sequentiality* for the second sentence—where we learn that John has cancer—was computed. We used the same, minimally informative topic ("a short story") to compute ensemble *Sequentiality* for both stories. As shown in Figure 3, including John in the first sentence significantly increases *Sequentiality* relative to the story where John is not mentioned in the first sentence. An independent samples t-test on *Sequentiality* values confirmed the statistical significance of the effect ($t_8 = 2.572$, $p = .033$).

We take this finding as a first piece of quantitative evidence supporting an information gain interpretation of *Sequentiality*. Without any reference to John in the first sentence, the second sentence has no relevance to the rest of the story, and therefore does not contribute any new information, leading to a lower overall *Sequentiality* value.

As a comparison, we computed an embedding-based analog of *Sequentiality* for each story type. This metric captures the semantic relatedness of each sentence, which we argue,

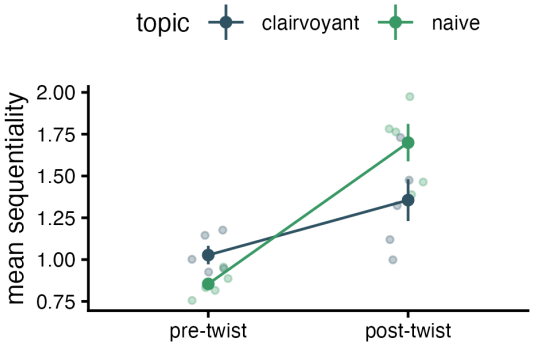


Figure 4: Ensemble *Sequentiality* values for annotations of *The Boyfriend*. Individual points are estimates from each model in the ensemble, averaged across sentences pre- and post-twist; error bars are standard error across these means.

given its sensitivity to text similarity, better represents narrative familiarity. We defined the topic term as the cosine similarity between the topic τ and the current sentence s_i. The contextual term is defined as the cosine similarity between one vector representing both the topic τ and the previous h sentences, and the current sentence s_i. Each story's overall score was computed as the mean of this embedding-based analog of *Sequentiality*. We used Google's Universal Sentence Encoder to generate and obtain the embeddings for each story. Figure 3 shows that this embedding-based metric is insensitive to the difference in contextual information between the stories, distinguishing *Sequentiality* from a metric capturing stimulus familiarity into one highlighting informational differences throughout the narrative.

Case Study 2: Contextual information gain in stories with "twist" endings

Previous work has shown that narratives with a "twist" or surprise ending induce marked changes in both the free recall and neural activity of humans engaging with the movie as they re-interpret the previously-observed events in light of the new information (Sava-Segal et al., 2026; Zadbood et al., 2022). Because this re-valuation process is directly relevant to the information gain hypothesis, we leveraged a movie in the FILMFEST dataset that has a twist ending—*The Boyfriend*—to further test our hypothesis.

We computed ensemble *Sequentiality* for annotations of *The Boyfriend* under two different topics with history $h = 8$. The "naive" topic was created to match the experience of a human watching the movie for the first time (topic = "A short movie titled *The Boyfriend*"), whereas the "clairvoyant" topic was created to match the experience of a human re-watching the movie after knowing about its contents (topic = "A short story about a robot woman replacing her robot boyfriend"). Annotations for the movie used to compute *Sequentiality* can be found in the Supplemental Materials.

Our information gain hypothesis predicts that the change in *Sequentiality* values after the reveal should be significantly greater for the naive topic relative to the clairvoyant topic, because each new sentence after the reveal includes more information about the new context that the story has shifted into (i.e., one with robots). Crucially, the familiarity hypothesis predicts the opposite effect: sentences that follow more predictably from the topic should have a higher *Sequentiality*, so sentences after the reveal should be greater for the clairvoyant topic relative to naive. Figure 4 shows that the data directly support our information gain interpretation over the original predictability interpretation: story topic and sentence position significantly interacted such that the average *Sequentiality* of events happening after the "twist" was significantly higher for the naive topic ($\beta = 0.5165$, $t_{12} = 4.306$, $p = .001$).

Case Study 3: Contextual information gain in nonlinear, long timescale narratives

Our final case study leverages a narrative custom-crafted by Chen and Bornstein (2024) to demonstrate the wooly, non-monotonic nature of credit assignment in the real world (Figure 5A). In the original story, a woman (K) demands a divorce from her husband (M). Finding himself in this undesired and unexpected state, M has to figure out which events in the past caused K to want to divorce him. If he simply "looked back" to the most recent negative experience with K (the purple event), M would incorrectly infer that failing to take out the trash is what caused K to demand a divorce. Alternatively, if he "looked back" to events where K was presented with new professional opportunities (blue arrows/events), he would incorrectly infer that K wants a divorce in order to advance her career. If somehow all M could think of were the neutral events colored in gray, he would be utterly confused about why K wants a divorce "out of nowhere".

We investigated (1) the within-story timecourse of *Sequentiality* as a function of topic and (2) how topic and sentence content interact to shape *Sequentiality* values. Figure 5B shows that the time course of ensemble *Sequentiality* values are qualitatively quite similar across topics. Figure 5C, however, shows how the average *Sequentiality* value for the different event/sentence types is modulated by story topic in a manner *not* predicted by a familiarity interpretation, but consistent with our contextual information gain hypothesis. Namely, event/sentence types that contain similar categories of information have *lower Sequentiality* on average than sentences that follow *un*predictably from the topic. The interaction term of a linear model reached statistical significance for the New Job topic x New Opportunity sentence interaction ($\beta = -0.7725$, $t_{26} = -12.60$, $p < .001$), as well as for the Naive topic x Neutral sentence interaction ($\beta = -0.2713$, $t_{26} = -4.43$, $p < .001$). Interestingly, the Divorce topic showed the least amount of variability by sentence type, perhaps reflecting the heterogeneous array of events that can cause divorces in real life.

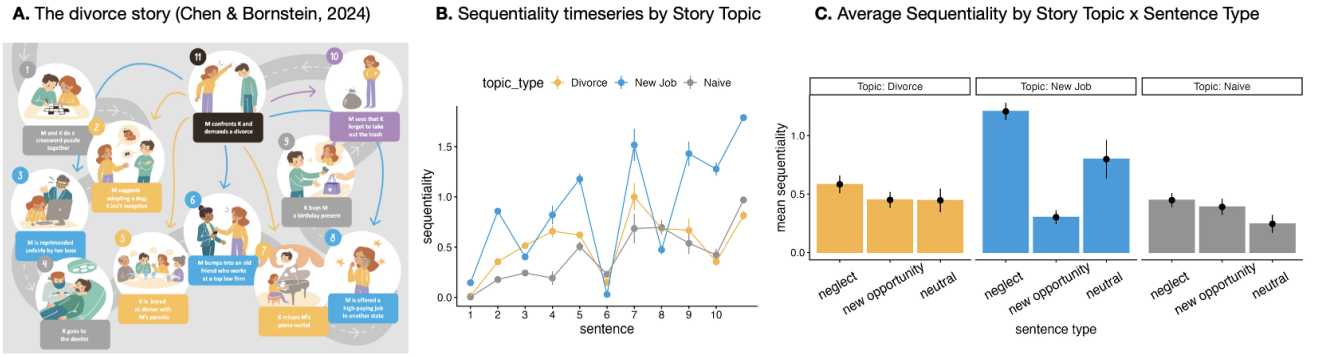


Figure 5: (A) Divorce story schematic adapted from Chen and Bornstein, 2024. Orange events correspond to moments of neglect, blue events correspond to moments indicating a potential opportunity, and gray events are neutral. (B) Time series of *Sequentiality* on the story across three different topics. Error bars are standard error across model type and history length (2,4,6,8). (C) Average *Sequentiality* for every topic across the three sentence types.

Results: *Sequentiality* tracks event importance and causal rank judgments

Having demonstrated that *Sequentiality* appears to track contextual information gain, rather than familiarity, we now provide empirical evidence linking *Sequentiality* to human ratings of event importance and causal centrality in the FILMFEST dataset (Lee & Chen, 2022). The stimuli in this dataset are 10 distinct short movies that were watched in succession by 21 human subjects undergoing functional magnetic resonance imaging. An independent sample of human subjects (n=4) rated the importance of each scene in each movie both during and after viewing. Additionally, a third sample of human subjects (n=12) generated cause-event pairings across all of the scenes in the FILMFEST movies. These ratings were subsequently used to generate *narrative networks*, where nodes correspond to events within a narrative and edges connect causally linked events (Lee & Chen, 2022). Edges are weighted according to the number of raters who identified the two events as causally related, and the causal centrality of each event can be quantified as the sum of these weighted edges. Lee and Chen (2022) additionally computed the semantic centrality of each event in FILMFEST by constructing semantic networks where the edge weights between events correspond to the similarity of those events in USE embedding space.

The FILMFEST database thus provides three distinct quantities that can be related to the *Sequentiality* metric: event importance, event causal centrality, and event semantic centrality. If *Sequentiality* is tracking how familiar or *similar over time* events in a narrative are, then we would expect it to be strongly related to event semantic centrality. Further, because the original *Sequentiality* metric was shown to decrease for narratives with greater numbers of major events (Sap et al., 2022), the original interpretation of *Sequentiality* predicts a null or even inverse relationship between event importance and event *Sequentiality*. However, if *Sequentiality* is instead tracking contextual information gain, then we would expect it to be *positively* related to event importance. The contextual information gain hypothesis makes no strong prediction about the association between semantic centrality and *Sequentiality*. However, it *does* predict that *Sequentiality* should be related to judgments of causal relations.

To answer these questions, we computed ensemble *Sequentiality* for each scene in each story in FILMFEST. We used a maximal history length, such that all scenes in the movie could fit into the context. Again, we constructed the naive and clairvoyant topics to mimic human observers watching the movie for the first time vs. rewatching. Naive topics were "A short story about 'Movie title'", and clairvoyant topics include the main event of the story. Clairvoyant topics for each story, along with the scene-level annotations used to generate *Sequentiality* values, are available in the Supplemental Materials. We z-scored *Sequentiality*, causal centrality, and semantic centrality within each story before inputting them into Spearman correlations.

In direct support of our information gain interpretation, Figure 6A shows that scene importance is significantly correlated with both "naive" and "clairvoyant" *Sequentiality* values ($r_{naive} = 0.36$, $p_{naive} < .001$; $r_{clairvoyant} = 0.23$, $p = .003$). Further, Figure 6B shows that the causal centrality of an event is also significantly correlated with "naive" and "clairvoyant" *Sequentiality* ($r_{naive} = 0.24$, $p_{naive} = .003$; $r_{clairvoyant} = 0.27$, $p = .001$). Finally, Figure 6C shows that the semantic centrality of an event has no association with either "naive" or "clairvoyant" *Sequentiality* ($r_{naive} = 0.01$, $p_{naive} = .894$; $r_{clairvoyant} = 0.02$, $p = .783$). This is in line with the case study (Figure 3) distinguishing metrics capturing event similarity, now across human judgments. Taken together, these results suggest that *Sequentiality* is better understood as tracking contextual information gain than schema-consistency of events in a narrative.

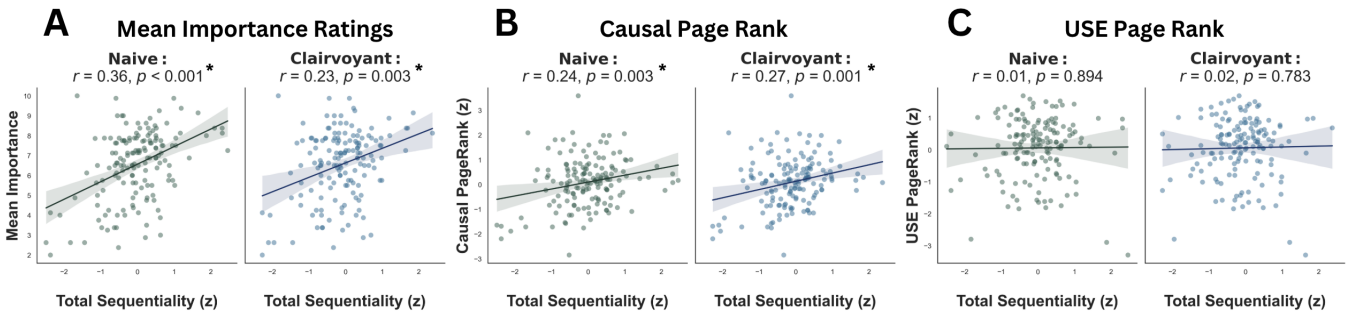


Figure 6: Spearman correlations between *Sequentiality* and FILMFEST measures of interest. (A) *Sequentiality* is significantly related to human ratings of event importance. (B) *Sequentiality* is significantly related to human judgments of an event's causal centrality. (C) *Sequentiality* is unrelated to an event's semantic centrality.

Discussion

We investigated the utility of the *Sequentiality* metric (Sap et al., 2020, 2022) for capturing the dynamic value of information as it unfolds while processing narratively-structured stimuli. This investigation is motivated by a recent theory that combines cognitive neuroscience findings with normative principles from reinforcement learning to propose that a functional role for narrative structure in human cognition: namely, facilitating efficient credit assignment by imbuing agents with complex, high-dimensional prior knowledge of contextual event transition probabilities (Chen et al., 2024). Testing this new theory requires methods for quantifying the value of information under different prior beliefs about narrative structure, a task we propose can be achieved by using the LLM-based *Sequentiality* metric. We combined formal derivations, proof-of-concept case studies, and direct application to human behavior to (1) test competing hypotheses about what cognitive construct the metric is best understood as capturing and (2) elucidate the role *topic* τ in shaping *Sequentiality* values. These two aims work together to enrich conceptual understanding of the metric, which is essential for calibrating its utility for studying credit assignment in humans.

Sequentiality was originally interpreted as reflecting how *schema-consistently* a piece of written text unfolds (Sap et al., 2022), implying that it reflects cognitive constructs related to familiarity: semantic similarity, fluency of processing, etc. If used as a metric for tracking the value of information—as suggested by Chen and Bornstein (2024)—this would suggest that narrative information is prioritized in the brain and behavior because of familiarity. Alternative accounts of event importance, however, suggest that *information gain* (or prediction error) is a primary construct driving the valuation of information in complex environments. To address aim (1), we first demonstrated that the original *Sequentiality* metric can be re-expressed such that it can be thought to quantify how much each observation (or sentence) s_i informs an agent about the time-evolving *context* within which they are processing information over and above what is expected given the topic τ (Equation 4).

The rest of the paper addresses aims (1) and (2) simultaneously. First, we presented a series of case studies on increasingly complex narrative stimuli to characterize how *Sequentiality* changes jointly as a function of story topic and event content (which can be consistent, inconsistent, or neutral relative to predictions of the model). Across narratives with radically different structures, we showed that *Sequentiality* is sensitive to the *mis*match between story topic and event content in a manner consistent with an information gain interpretation. Namely, *Sequentiality* for sentences that contain topic-consistent information are systematically *attenuated* relative to sentences that deviate in content from the topic (Figures 4 and 5). Finally, we showed that *Sequentiality* values computed for narrative stimuli in the FILMFEST dataset significantly correlate with human ratings of event importance and causal centrality, but *not* semantic similarity (Figure 6). Taken together, these converging findings strongly suggest that *Sequentiality* captures a quantity more similar contextual information gain than familiarity or fluency of processing.

While promising, our findings are limited in important ways. First, the narrative stimuli used in the case study reflect a small sample of the space of possible narratives that the metric can be applied to, which may raise concerns about the generalizability of our findings. Future work is needed to more comprehensively investigate the information gain hypothesis across narrative types. Second, the evolving capabilities of LLMs may change the trends we observe as they are updated moving forward. To help remediate this, we designed our implementation so that the specific models being used to calculate probabilities can be easily swapped, and designed the ensemble metric to mitigate the effects of individual models and increase the validity of the metric.

How we structure and coherently piece together experiences modulates the information we receive from them. Designing a model that can measure how well a narrative captures the level of information humans receive allows us to automatically identify narrative structures that humans can use to learn and transmit information. Future work will identify if *Sequentiality* segments stories into events, where shifting contexts leads to temporary dips in information gain and in-

Sap, M., Jafarpour, A., Choi, Y., Smith, N. A., & Horvitz, E. (2022). Quantifying the narrative flow of imagined versus autobiographical stories. *Proceedings of the National Academy of Sciences of the United States of America*, *119*(45), e2211715119. https://doi.org/10.1073/pnas.2211715119.

Sava-Segal, C., Grall, C., & Finn, E. S. (2026). Narrative "twist" shifts within-individual neural representations of dissociable story features. *Proc. Natl. Acad. Sci. U. S. A.*, *123*(11), e2512071123.

Van Hasselt, H., Madjiheurem, S., Hessel, M., Silver, D., Barreto, A., & Borsa, D. (2021). Expected eligibility traces. *Proc. Conf. AAAI Artif. Intell.*, *35*(11), 9997–10005.

Zadbood, A., Nastase, S., Chen, J., Norman, K. A., & Hasson, U. (2022). Neural representations of naturalistic events are updated as our understanding of the past changes. *Elife*, *11*(e79045).

Supplemental Materials: Quantifying information gain in narrative stimuli using open-source large language models

USE Sequentiality on HIPPOCORPUS

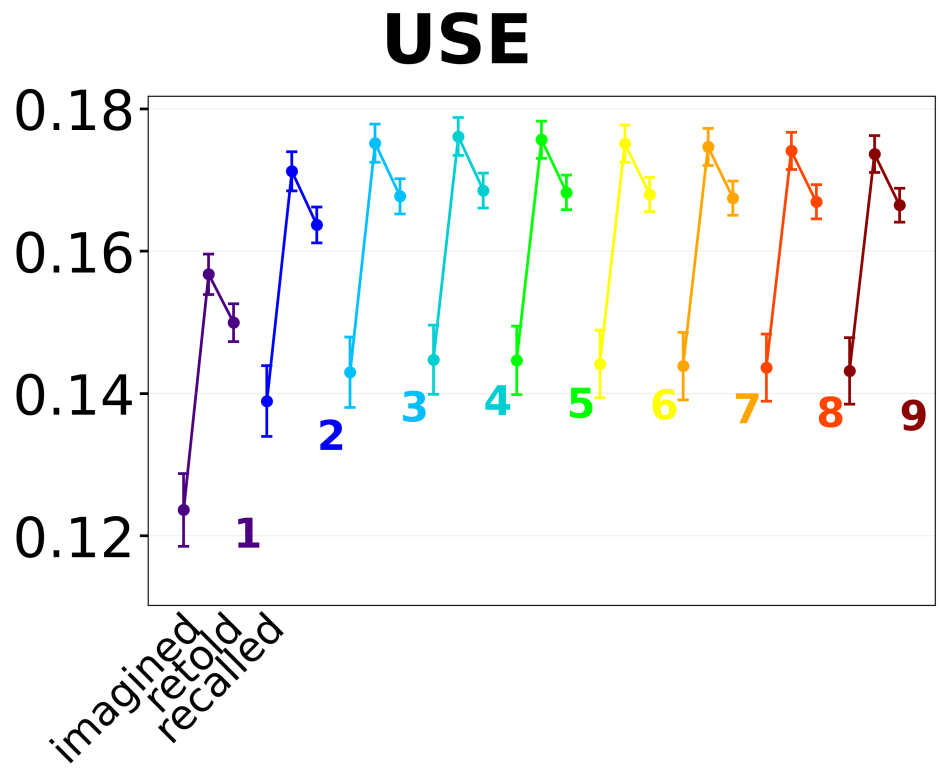


Figure S1: We calculated Sequentiality values using Google's Universal Sentence Encoder (USE) rather than a large language model to offer an alternative definition of narrative flow. Rather than focusing on the probability of a sentence given what has come before, this interpretation uses measures of textual similarity. The metric generates three embeddings: one that embeds the topic and the previous *h* sentences to establish a context, one that just embeds the topic, and one that embeds the current sentence. Then, the cosine similarity between the current sentence and the other two vectors is calculated, and the mean between those similarity values is calculated. This mean is what is plotted above. USE Sequentiality does not follow the same trends as its LLM-based counterpart. USE Sequentiality does not increase monotonically as a function of history length, nor is the relative ordering of the story types preserved (imagined no longer has the highest Sequentiality value out of all of the story types).

Wedding Story Case Study

We aimed to investigate the effects of shuffling the order of the sentences in a narrative on their Sequentiality values. To do this, we considered a subset of HIPPOCORPUS. We filtered the dataset by topic and only included stories with the topic "my wedding." There are only three stories in the dataset with that specific topic: one from each story type.

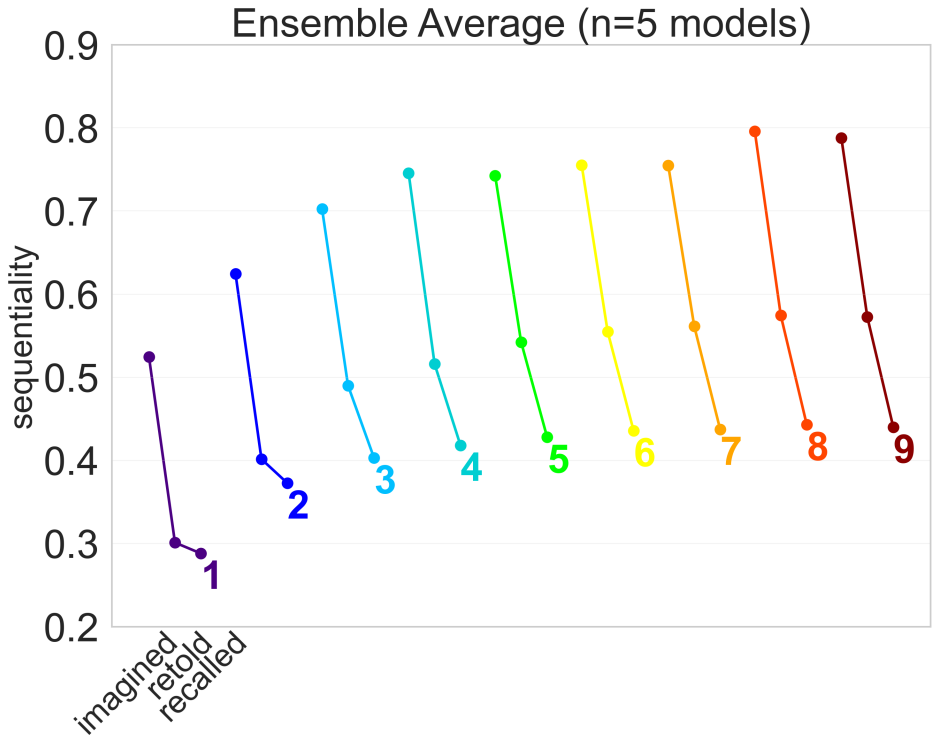


Figure S2: The Sequentiality values of the three stories in the subset follow the same trends as the larger dataset. All Sequentiality values increase monotonically as a function of history length, and the relative ordering of story types is preserved. The imagined story has a much higher Sequentiality value than the retold and recalled one.

Next, each story was shuffled 100 times, creating a dataset of 303 stories in total (300 shuffled + 3 original). Additionally, we calculated Sequentiality values for six topics in total where three were from other stories in the dataset, one was the opposite ("my divorce." instead of "my wedding."), and one was a control ("A short story").

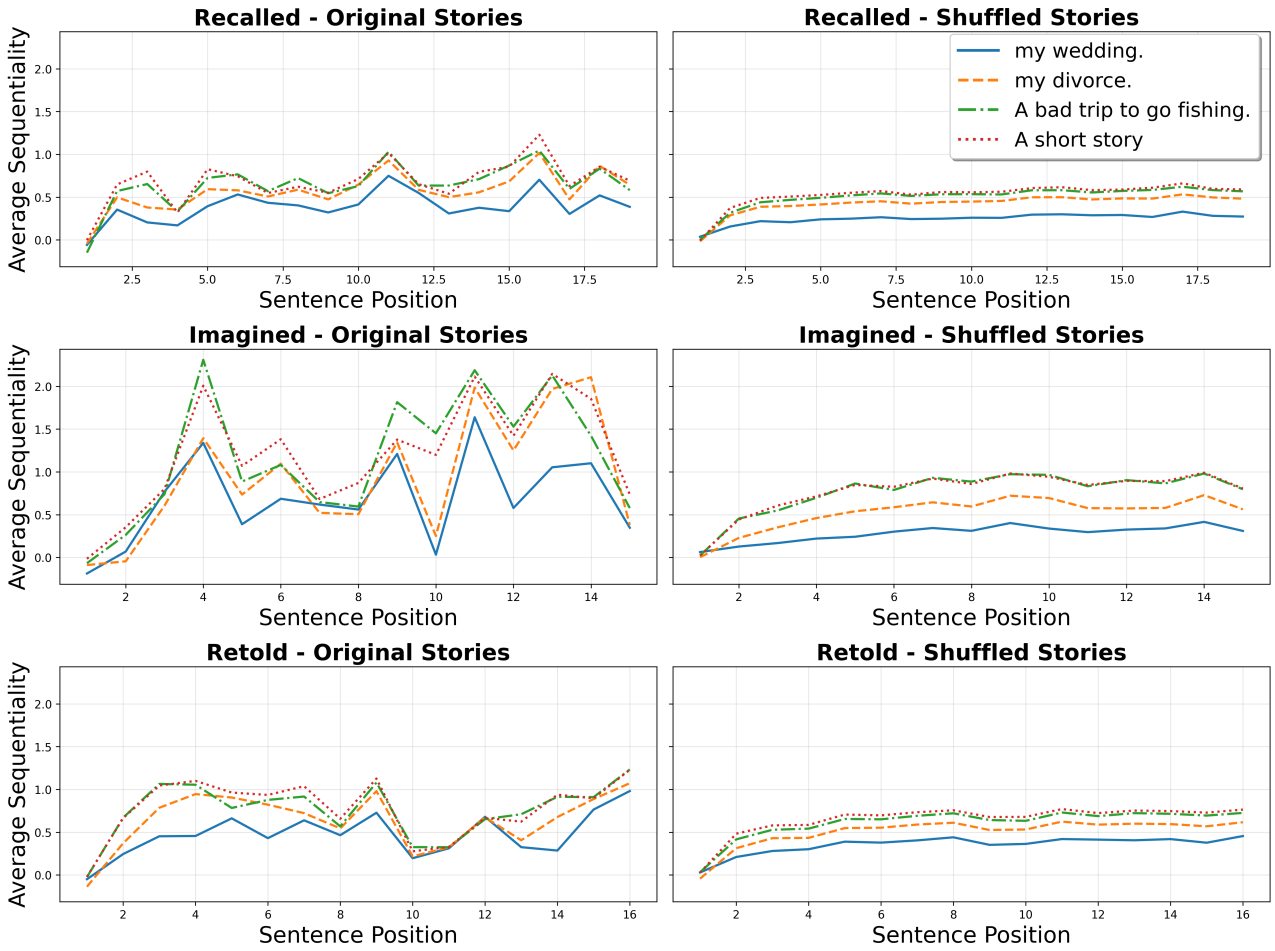


Figure S3: Sequentiality was calculated for the same three stories for six different topics. Each row in the figure corresponds to a story type. The left column plots Sequentiality values for unshuffled stories across six topics, and the right column plots Sequentiality values for shuffled stories across six topics. Among topics in the shuffled condition, the actual topic has the smallest Sequentiality value throughout the timeseries. Since the original topic is the most informative, it provides the most information about what may happen in each sentence, even when shuffling their order. Since the model already has this information, there is less gained overall, leading to the lower value. Additionally, shuffled stories are much less volatile than their unshuffled counterparts, suggesting that Sequentiality is sensitive to the ordering of the sentences in a narrative.

Temperature Analysis

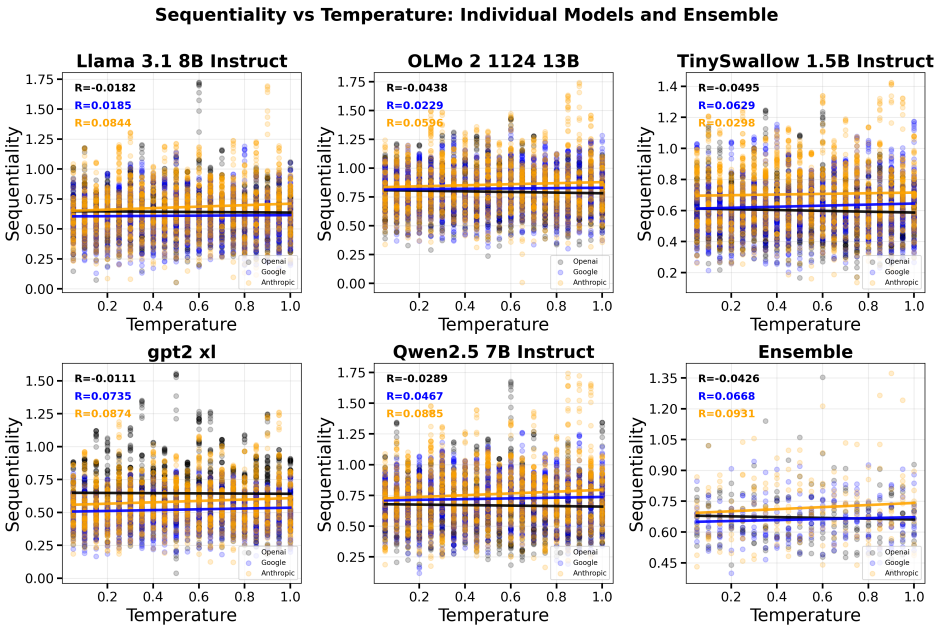


Figure S4: To test whether model temperature influenced Sequentiality, we generated 600 stories across three model providers. Specifically, we generated stories based on 10 topics (seen below) on each model at 20 different history lengths in the range $[0, 1]$ at increments of 0.05. We used GPT 5.2 from OpenAI, Gemini 3 Flash from Google, and Claude Sonnet 4.5 from Anthropic. Sampling from multiple model providers mitigates any bias that any singular LLM may have picked up from its training data. Each LLM was given the same prompt to generate a story ("Write a short story between 7-13 sentences long about the following topic: story. Do not include any metacommentary, only return the story."). The thought was that a more stochastically generated story may be less predictable and therefore we would expect a lower Sequentiality value under narrative flow interpretation of the metric. We ran linear regressions for each text-generating model. We can see in the plot that across all models in the ensemble and the ensemble itself that there is no significant correlation between model temperature and Sequentiality values ($R < 0.1$ for all generation models).

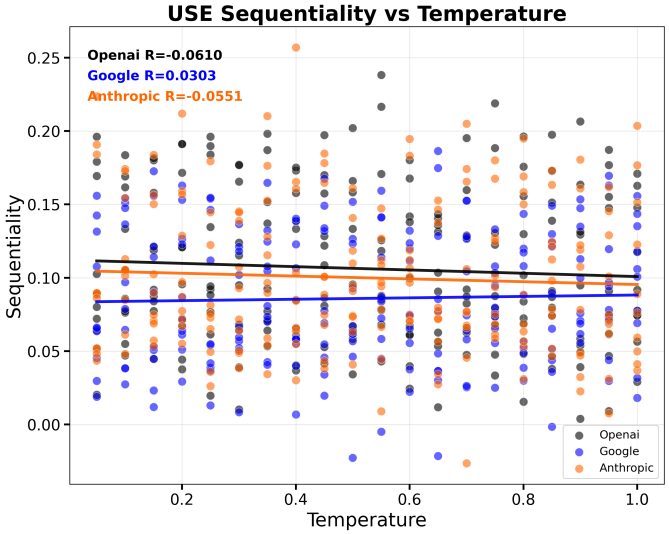


Figure S5: USE Sequentiality is temperature invariant. We found that similar to LLM-based Sequentiality, the temperature with which a model is generated how no effect on it's USE Sequentiality value across model providers.

Below are the topics that were used to generate stories for the temperature analysis:

```
story_prompts = [
    "Old friend calls after twenty years of silence.",
    "You find someone's wallet and recognize the photo inside.",
    "Neighbor asks you to keep a strange secret.",
    "You overhear a conversation not meant for you.",
    "Unexpected letter arrives with no return address or name.",
    "A chance meeting changes your entire daily routine.",
    "You help a stranger and they never say thanks.",
    "Lost phone reveals someone's hidden double life.",
    "You see someone you know pretending not to notice.",
    "Childhood home is up for sale again."
]
```

Divorce Story across Different Endings

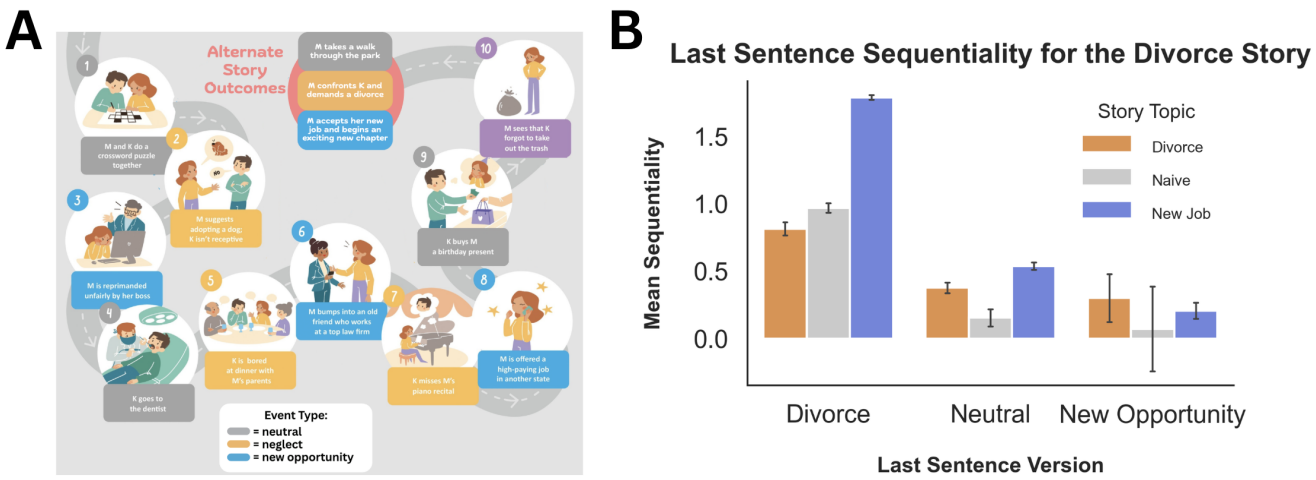


Figure S6: Last sentence sequentiality of the Divorce Story across multiple endings. To assess how changes to the ultimate outcome relate to changes in measured information gain, we created two new final sentences in the story (A). The divorce version maintains the same sentence as the original version, the naive ending is "M takes a walk through the park", and the new opportunity ending is "M accepts her new job and begins an exciting new chapter". We only looked at the last sentence since the Sequentiality values for the other sentences were identical. Error bars are standard error across history lengths (2,4,6,8) and LLM. (B) There was a significant interaction between the neutral last sentence outcome and new job topic ($\beta = -0.8133$, $t_26 = -3.528$, $p = .002$) along with a significant interaction between the new opportunity sentence outcome and the new job topic ($\beta = -1.0678$, $t_26 = -4.632$, $p < .001$).

Topics in HIPPOCORPUS subset

Table S1: Main Events by Frequency

Main Event	Occurrences
about my cousin's wedding	9
my wedding	7
getting a promotion	7
a wedding	6
my cousin's wedding	5
my friend's wedding.	5

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
my surprise birthday party.	4
My wedding	4
graduating from college	4
my dog dying	4
getting a promotion at work	4
Family vacation	4
about getting a promotion	4
a birthday party	4
about my wedding	4
the birth of my son	4
my wedding day.	4
my cat dying.	3
Hard work is the foundation of success	3
A car accident	3
my sister's wedding.	3
going on vacation	3
a car accident	3
the death of my mother	3
The birth of my first child	3
the birth of my daughter	3
my sister's wedding	3
having a baby	3
the death of my uncle	3
my brother's wedding	3
my cousin's wedding.	3
my birthday party.	3
my daughter coming out	3
about failing to win a competition	3
The birth of my daughter.	3
getting a new job	3
my grandmother's passing	3
losing my job.	3
my wedding.	3
my son's first birthday party.	3
my child being born.	3
the birth of my child.	3
My cousin's wedding	3
breaking up with my boyfriend.	3
adopting a dog	3
A family reunion	3
a birthday party.	2
about falling in love	2
my friend and I going shopping for a dress.	2
an autistic adult's social skills	2
reconnecting with family.	2
The birth of my nephew	2
My son's HS graduation	2
starting a business.	2
My insane landlord	2
my pants splitting in public.	2
family vacation	2
bumping into a childhood friend	2
finding out about the death of a friend	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
my dad's death.	2
Me quitting drinking.	2
seeing my recently born nephew for the first time.	2
about a family in need of help	2
closing the family estate	2
birthday celebration tribute	2
Our picnic trip turned out to be a Panic trip	2
Hurricane prepping	2
my nephew's nap time.	2
About a trip to a new building that my company opened.	2
25th Wedding Anniversary trip to French Polynesia.	2
trying to sell my house before it went into foreclosure	2
my daughters court of honor	2
the first day of vacation.	2
A wedding that we attended	2
About visiting the zoo on my son's birthday	2
How I failed to rescue a dog from being hit by cars.	2
my son's hearing test	2
giving birth to my son.	2
engagement	2
attending the college graduation of my best friend's son.	2
about the birth of my daughter	2
Mother's day lunch	2
falling in love	2
about my car accident incident	2
the birth of my first child	2
interviewing for a new job	2
A family party.	2
My youngest child starting kindergarten.	2
about a car accident I survived	2
a dinner to impress guests going wrong.	2
my mother's cancer	2
Breaking up with my long time girlfriend.	2
meeting a friend for the first time	2
my father's death.	2
kids going to college	2
Love	2
my child losing a child.	2
Our family was moving to a hobby farm.	2
me getting a promotion.	2
me starting my own podcast	2
my mom's memorial service.	2
the time I had at my sister's wedding	2
a dog dying	2
My wedding day	2
giving birth.	2
meeting an online friend for the first time.	2
About my softball team playing a quadruple header	2
my graduation ceremony.	2
The story is about a wedding I attended recently.	2
about my grandfather's funeral	2
my feelings on forgiving my mom	2
Having my dog euthanized.	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
my nephew's birthday party	2
About moving	2
a fight with my landlord	2
Reconnecting with an old friend.	2
My health scare and how it changed my outlook on life.	2
the funeral of my grandmother	2
celebrating birthdays by going to an amusement park.	2
my son leaving to move into his first apartment	2
about my wife and I reflecting on our time together.	2
a woman falls in love after her heart is broken.	2
my daughter getting married and moving away	2
my sister happyness	2
Birthday party	2
About my sister's wedding	2
a bucket list float trip	2
Going on a date for the first time in a couple years.	2
my stepfather passing away.	2
About getting a new dog	2
first grandchild birth	2
going to a concert	2
my promotion	2
selling my beloved car	2
an event at school leading to a promotion	2
My cat passing away	2
my family reunion	2
my grandmother's passing and funeral.	2
my house burning down.	2
an intruder in my parents home.	2
The day I was patched into my motorcycle club	2
about my daughter 14th birthday	2
Taking a vacation.	2
a street mishap	2
about going to marthas vineyard with friends	2
my friend's hard time	2
about my best friend's son.	2
Getting a raise.	2
my Medical Lab Tech education.	2
my second daughter taking her first steps.	2
my mother's thyroid cancer consultation	2
About a health scare	2
my experience with walking meditation.	2
Hiking to the top of Mt. Leconte	2
Being proposed to for the first time.	2
proposing to my girlfriend.	2
my grandfather's death	2
about my personal best workout	2
my friend passing away very unexpectedly.	2
Going to see a concert.	2
a family trip to Sesame Place.	2
My husband and I going on vacation	2
My feelings about sending my child off to live on their own.	2
my sister's death	2
purchasing my dream horse.	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
needing help financially by couponing	2
my kids leaving for college	2
We were given a discount, and then asked to give it back	2
my Dad's funeral	2
my best friend visiting for my birthday.	2
my husbands job loss	2
A breakup	2
my niece starting kindergarten	2
Winning an election	2
rescuing an abandoned ferret.	2
my son's Eagle Scout ceremony.	2
proposing to my girlfriend on the beach.	2
my mother coming to visit.	2
Me becoming an aunt.	2
Rekindling a relationship.	2
a birthday party spent with a family member.	2
promotion at work	2
losing the opportunity to see my nephew.	2
losing a job that I had for nearly twenty years.	2
booking my first wrestling match	2
discovering a new relative.	2
buying our first home.	2
About a close relative being diagnosed with terminal cancer.	2
me and fiance moving in together and buying a house.	2
feeding giraffes at the zoo.	2
my aunt dying	2
a family reunion	2
my experience camping for the first time.	2
The death of my dog.	2
winning a seemingly impossible competition!	2
Love at my Cousin's Birthday	2
My father passing away and its effect on me.	2
the process of purchasing my first home.	2
parent quitting smoking	2
a dinner	2
a vacation that I took with my sister.	2
about a close friend's wedding.	2
Me and boyfriends relationship	2
going on a tough hike in the mountains.	2
A cruise	2
family celebration	2
my sons first birthday party	2
how i met my boyfriend.	2
about getting married.	2
Finding our my sister is pregnant.	2
Me, watching a baseball game.	2
the last time I saw my brother before he passed away.	2
A job change	2
attending a friend's baby shower.	2
my cat's blocked bladder	2
My mother's passing.	2
about my engagement.	2
my husband's poor health.	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
my first day in Tae Kwon Do class.	2
my sister in law having my nephew.	2
A trip to California for my fathers 80th birthday.	2
the death of my dog.	2
a first date with someone i met online	2
A relationship and then breakup	2
An emergency trip to the hospital.	2
experiencing a mass at Notre Dame in Paris.	2
having my first child and being a daughter.	2
My girlfriend and I moving in together.	2
getting a puppy	2
Moving into our new house	2
a day with family.	2
Working in the Florida Keys	2
a trip to Denver to buy cannabis edibles	2
a new addition to the family	2
CHILD BIRTH	2
about presenting my real self to a stranger	2
starting a relationship online.	2
becoming a family of five.	2
a proposal.	2
A vacation to my cousins house on 4th of July.	2
my college graduation	2
a therapy dog named Sheriff and a girl named Rebecca	2
a garden giveaway event for low income people	2
I found out new information about the night my uncle died.	2
Our first family vacation to Walt Disney World.	2
when I got married.	2
my daughter's wedding	2
teaching a lesson to a class of seventh graders.	2
my best friend and boyfriend meeting for the first time.	2
A trip to a convention in Baltimore with some friends.	2
About an act of kindness between total strangers.	2
my nephew's graduation.	2
a project I worked on at work	2
my baby niece getting born	2
I thought my dog was very ill.	2
Daughter playing softball	2
my boyfriend and I visiting Denver for the third time.	2
a family event that I went to.	2
Losing my dog	2
quitting my job.	2
winning a competition.	2
My brothers wedding	2
Wedding	2
about my brothers wedding	2
working as a nurse	2
Grand child experience the 4th of July for the 1st time.	2
an incredible day on our vacation in Boulder, Colorado.	2
a flight getting canceled	2
finding a job after being unemployed for 6 months.	2
how my relationship started.	2
moving	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
a presidential rally	2
Me losing my job	2
giving birth to my son	2
My parent's birthday party	2
a road trip I took after graduation.	2
about the vacation trip	2
staying calm in the face of pressure	2
facing my fears at a water park.	2
Hiking in Georgia	2
About my best friend's wedding	2
my birthday party	2
About my portrait being published	2
the death of my uncle.	2
Being proposed to while on vacation.	2
The improving relationship between my daughter and myself	2
taking part in a Haiti charity dance event.	2
Death in family	2
my neighbor's funeral.	2
falling in love with a friend	2
Death of a dream	2
going to a baseball game.	2
how my cat helped my marriage	2
the death of a family member.	2
a person close to me performing in concerts.	2
Losing a job	2
The birth of my second child	2
my partners Type 2 diabetes diagnosis and lifestyle changes	2
My surprise birthday party	2
dropping off my oldest child to college	2
my trip to Ireland and New York City.	2
waking up to my dog being paralyzed.	2
the death of a good friend's mother	2
my son's graduation from high school	2
Taking a trip despite travel anxiety.	2
My wife's birthday	2
about my wedding day.	2
About an acccident	2
me reuniting with a bully.	2
my sister-in-law who died recently.	2
a kid being hit by a car.	2
My father passing away	2
Seeing a UFO	2
studying foraging.	2
my son's coming home party.	2
my graduation.	2
My home sale	2
almost killing someone	2
Moving away from my girlfriend.	2
my mother dying	2
a promotion at work	2
Going to Disney World.	2
a family vacation.	2
buying a new house	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
the passing of my dad.	2
going to a friend's funeral	2
betrayal from a best friend.	2
giving swimming another try after quitting it	2
family relatives with Bi-polar/Depression go through	2
an unexpected job change.	2
waiting for a proposal.	2
My best birthday	2
buying a house	2
about my friend's wedding.	2
Getting a promotion	2
Reconnecting with a former love	2
the death of a friend.	2
About my Nephews Wedding	2
Taking my son to college.	2
setting up a backyard movie theater for my daughter.	2
my first time moving in with a romantic partner	2
a job promotion	2
getting a job	2
When I graduated from college	2
going back to college	2
having a Chinese dinner with my brother and my family.	2
my mother's funeral.	2
a family friend's birthday party	2
my brother's wedding.	2
spending time together with my family	2
a family vacation on a Mediterranean cruise.	2
my PTSD diagnosis.	2
my first visit to meet my girlfriend's family.	2
Filing for Bankruptcy	2
my first broken bone ever, from a skiing accident.	2
About traveling to Yogyakarta Indonesia	2
my nephew being born.	2
My physical and mental health deteriorating.	2
my kitchen remodel	2
Decision to have eye surgery.	2
the birth of my fourth child.	2
My dog passing away.	2
My son leaving for a job in a different state.	2
A family trip.	2
a day of sightseeing in Brisbane with my mom & myself	2
finding money on the street.	2
A Birthday Party	2
the release of my first book.	2
a white water rafting trip I took with my work.	2
my mom having a heart attack.	2
a car accident.	2
an immediate family member's perceived illness.	2
surgery for gallbladder removal.	2
winning custody of my children	2
Rescuing a kitten	2
A birthday party	2
my daughter learning to crawl	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	**Occurrences**
the birth of my son.	2
going to Florida for vacation.	2
spending time with my family	2
A memorable fourth of July.	2
our first family vacation	2
my son's graduation.	2
having a second child and the emotions im going through	2
taking son to college.	2
a trip to Lassen National Park with my brother.	2
camping with my sons.	2
first independent swim lessons	2
my feelings on the day of my fathers funeral	2
teaching my daughter to drive.	2
my ex boyfriend's death.	2
an engagement party.	2
Me proposing to my now wife	2
going out to a live show	2
my prostate cancer	2
about finding out I am going to be a father	2
my mother-in-laws 60th birthday.	2
My aunt's sudden death and my job as her executor.	2
about nephew graduating high school	2
My husband's journey to satisfying employment	2
my friend's wedding dress appointment	2
my first birthday celebration with my college friends.	2
meeting a friend.	2
my Parkinsons diagnosis	2
My dog having some health problems.	2
getting accepted to a genetics PhD program	2
My anniversary weekend	2
being attacked by a homeless man camped out at our office.	2
my first trip to Vietnam	2
my second heart attack	2
about a life or death situation that happened to my mother.	2
A visit from our cousins.	2
Saving a man's life.	2
my promotion in a business after switching careers.	2
the death of my mother.	2
My brother's wedding	2
a friends accident	2
about my failed company	2
being homeless	2
My best friend's wedding and I ended up engaged the same day	2
About getting a promotion.	2
starting a business	2
my uncle's funeral	2
How I met my boyfriend.	2
Visiting a foster child in a rehabilitation facility.	2
getting engaged at a concert	2
the birth of my baby girl.	2
me getting fired from my job	2
the death and funeral of my best friend.	2
going to a concert.	2

Continued on next page

Table S1 – *Continued from previous page*

Main Event	Occurrences
a shooting while my kids played outside	2
my wedding anniversary	2
my son's first day of Kindergarten.	2
dropping my toddler off at nursery school	2
making a trip to six flags new england.	2
My stepson coming to live with us.	2
a long overdue family reunion	2
an emotional encounter with a previous patient	2
My depressed teenage daughter.	2
about surgery and complications which developed	2
getting into an accident after a great time at a performance	2
mine and my children's birthday party	2
Buying our first house.	2
my summer vacation	2
my music duo splitting and our respective future plans.	2
About getting a promotion	2
going to a wedding	2
a class reunion which brought me close to ex girl friend.	2
my first time on a roller coaster.	2
quitting a job and starting a new one.	2
I was sick one night and my significant other finally cared!	2
me and my dog.	2
my daughters first day of school	2

Film Fest

Table S2: Summary of movie stimuli from FilmFest dataset

Movie	Num Sent.	Naive Topic	Clairvoyant Topic
CMIYC	18	A short story titled 'Catch me if you Can'	A short story about a famous con man being captured
Record	18	A short story titled 'The Record'	A short story about a woman rapidly aging through her entire life
Boyfriend	18	A short story titled 'The Boyfriend'	A short story about a robot woman replacing her robot boyfriend
Shoe	20	A short story titled 'The Shoe'	A short story about a man and a woman mimicking each other until he is hit by a car
KeithReynolds	15	A short story titled 'Keith Reynolds'	A short story about a man hoping for a promotion who ends up killing a coworker
Rock	19	A short story titled 'The Rock'	A short story about a mountain becoming space debris
Prisoner	18	A short story titled 'The Prisoner'	A short story about a man trapped in a strange village
BlackHole	18	A short story titled 'The Black Hole'	A short story about a man getting trapped in a safe
Post-it	16	A short story titled 'Post-it Love'	A short story about two coworkers creating a mural of themselves
BusStop	19	A short story titled 'Bus Stop'	A short story about a man at a bus stop that turns into a dog

Film Annotations

CMIYC:

A television game show host welcomed the audience.
The host announces three men to the stage, each claiming to be Frank William Abagnale.
The host described a series of elaborate impersonations committed by the real Frank William Abagnale.
He emphasized that these crimes occurred before Abagnale turned nineteen.
The host explains that the audience can ask these three men any question.
A panelist questioned one man about why he chose crime over honest work.
He explained that money and opportunity motivated his actions.
Another panelist asked the second man who eventually captured him.
The second man said it was Carl Hanratty.
The scene shifted to a rainy prison in France.
An FBI agent named Carl Hanratty arrived to extradite Abagnale.
He read legal documents outside Frank William Abagnale's prison cell.
Inside, Frank coughed and pleaded for help.
Carl ignored Frank's pleads and continued to read the documents.
Frank collapsed, and Carl demands a doctor.
Carl explains that he needs Frank back in America for a play, but the guards ignore him.
Frank disappears from the room in an effort to escape.
Carl and the guards catch Frank, and Frank agrees he is ready to go back to America.

Record:

A woman stood alone in a room with a pizza box.
She opened the box and took out a slice.
A knock sounded at the door.
She opened it and found a record.
She placed the record on her player and lowered the needle.
Cheerful music began to play.
She swayed to the music and while lifting the pizza the record skipped, and part of the pizza vanished.
She noticed the change, stopped the record, and spun it backward.
The missing pizza returned to her hand.
She experimented by spinning the record back and forth and the pizza progressively vanished and reappeared.
She advances the record further, and gets teleported to a new house.
She is now older, both married and pregnant.
She spins the record and notices her pregnant belly grows and shrinks.
She advances the record some more and has her baby in her hand.
The baby starts crying, so she rewinds the record only to end up as a child now.
She struggles to reach the record, and advances it quite a bit and now is an elderly woman in a wheelchair.
She can no longer reach the record without time resetting her, so she falls over and the record goes even further.
She is now even more elderly, and as she walks over to the record player the record ends and only her ash jar remains.

Boyfriend:

A woman and man sat across from each other while quietly eating dinner.
The man ate his dinner without looking up.
The woman poured wine for him, but he politely refused.
The woman reminded him it was their anniversary, but he kept eating.
The man commented on the asparagus, and she teased him about its aphrodisiac qualities.
The woman grew frustrated and scolded him for not celebrating their anniversary.
The argument escalated, and she threw her wine at him in anger.
The man ignored her outburst and lit a cigarette.
The woman snatched the cigarette from his mouth and threw it aside.
She reaches behind his neck to turn off a switch and make him power off, revealing he was a robot.
The woman moves to her computer, scrolls through websites displaying robot men, and ordered one to arrive in his place.
The robot man was delivered, wrapped in plastic, matching her specifications perfectly.

She removed the ring from the original man and placed it on the robot's hand.
The robot came to life, awakening with mechanical precision.
The robot massaged her shoulders gently, mimicking the original man's gestures.
The woman smiled, satisfied, as her new partner followed her every command.
The robot reaches toward the woman's neck to reveal a switch.
The robot turns off the switch and the woman goes limp; she was a robot too.

Shoe:

A dirty black running shoe lay on a sidewalk.
The shoe turned upright as a foot tapped the ground.
A candy wrapper fell beside the shoe.
A milk carton was opened.
A man looked down and smiled slightly.
He licked his lips.
He peered into the carton and took a drink.
He glanced around and drank again.
Across the street, a woman walked in a red shirt and black skirt.
The man leaned against a maroon car.
Another man approached him in a confrontational way.
The man with the milk backed away.
He walked down the street, watching the woman.
He tripped on the curb but recovered.
The woman looked toward him, then stumbled while walking.
They began copying each others movements.
Their gestures grew playful and exaggerated.
The man stepped forward without noticing traffic.
A car struck him, sending his car and shoes flying through the air.
One of the shoes lands next to another black shoe.

KeithReynolds:

Keith stepped out of his car and walked toward the entrance.
A narrator explained that this was Keith Reynolds on promotion day.
The narrator described him as the most senior junior business analyst, fully expecting to get promoted.
On the fourth floor, he noticed Sarah, a girl he has a crush on, near a file cabinet.
Keith offered her a cup of coffee, which she did not want.
She poured it out the window, injuring a man outside.
Keith goes to see his boss but does not get the promotion.
He later learned the promotion had gone to Patrick Campbell instead.
Immediately after, he saw Patrick and Sarah kissing together.
Anger grew as Keith confronted Patrick and ended up killing him.
Keith attempted to drag the dead body down the stairs.
Another coworker saw this and fainted, falling down the stairs and dying as well.
Keith attempts to take the body down a ladder, but Sarah sees him carrying Patrick's dead body.
Keith imagines her jumping into his arms and loving him, but instead she faints as well.
Keith finally escaped through a window, hanging from a ladder as his former life faded behind him.

Rock:

A living mountain appeared alongside ancient underwater creatures and a bright sky.
Small ancient creatures moved around as the mountain blinked and eventually fell asleep.
While it dreamed, time accelerated and landscapes rapidly transformed.
Dinosaurs ran past as the mountains body changed shape.
Ice and snow covered the land as glaciers pushed the mountain into a boulder.
The climate warmed, forests grew, and the boulder rested beneath a tree.
Human activity awakened the boulder as men chopped wood and built nearby homes.
Disturbed, the boulder uprooted itself and wandered until it was captured by people.
The boulder was carved, reshaped, and put to work as machinery.
Time sped forward as it became a millstone and then a wheel.
Soldiers seized it, crushed it down, and fired it as a cannonball.
The shattered stone sank into the sea and slowly washed ashore.
Now small, the rock moved through a modern world of farms and roads.

Scientists discovered it and placed it in a museum.
An accident sent it tumbling into the streets, where it was broken apart again.
Reduced to dust, it was swept away and transformed into glass.
That glass became part of a telescope launched into space.
The spacecraft collided with a planet and melted into the surface.
Over time, a new mountain formed on that planet and fell asleep once more.

Prisoner:

A man in a black suit approached a maid cleaning a table.
He asked where he was, but her answers were vague.
Growing anxious, he asked for the police station.
She explained that there was none.
He asked to use a phone and was directed to a phone box.
Ominous music played as he walked through the village.
At the phone box, he tried to make a call.
The operator refused him because he had no number.
The man walked back through the street, scanning his surroundings.
He stopped at an information board and pressed a button.
A taxi appeared almost immediately.
The driver spoke casually as she drove him through the village.
They passed houses, fountains, and narrow roads.
She dropped him off near a white building and drove away.
The man entered a general store and asked for a map.
The clerk offered only maps of the village.
Every map showed the village enclosed by mountains and named Your Village.
Realizing he could not leave, the man exited the store.

BlackHole:

A tired man stood beside a copy machine and sighed.
He pressed a button, and the machine beeped repeatedly.
Annoyed, he kicked the copier.
The machine printed a solid black circle on white paper.
The man picked up the page and studied it.
He set it aside and checked if anything was in the copier but there was nothing.
He reached for his drink and placed it on the paper.
The cup vanished into the black circle.
Startled, the man leaned closer to inspect it.
He cautiously tested it with his hand.
To his surprise, he pulled his cup back out.
Curious, he experimented by retrieving candy from a vending machine.
The man smiled, encouraged by his discovery.
He noticed a locked door marked Keep Out.
Using the black hole, he unlocked the door from the inside.
He placed the paper on a safe and pulled out cash.
Greed took over as he climbed fully into the hole.
The paper fell away, leaving him trapped in the locked safe.

Post-it:

A woman stood at a printer, quietly making copies.
Across a thin border, a man did the same.
Both stood up at the same moment and noticed each other.
They quickly looked away, flustered and smiling, and returned to work.
The man sat at his desk and opened a blue folder.
Inside, he found a smiley face made of pink post-it notes.
He looked around, confused but amused.
The woman briefly peeked out from her cubicle and hid again.
The man smiled and thought for a moment.
At her desk, the woman opened the blinds.
A sun made of post-it notes filled her window.
She smiled happily and sat down.

Later, the man adjusted his tie in the bathroom mirror and noticed a heart made from post-it notes.
The woman gathered armfuls of post-it notes from a supply closet and hurried down the hallway
She saw the man making a giant post-it note mural of their faces on the wall.
The man turned to her, and they smiled at each other.

BusStop:

A man waited at a bus stop, sweating through a red T-shirt, holding a plastic grocery bag with melting ice cream inside.
A black SUV screeched to a stop nearby.
A frantic woman rushed to a payphone, failed to make a call, and then begged to borrow the mans phone.
She called someone on it, argued angrily into it, and smashed it onto the street.
The man protested and asked for some money for the phone as she broke down crying claiming another man is on the way.
Another man arrived under a bridge, blasting loud music.
The woman confronted him furiously, accusing him and demanding to know where someone was.
After the new man demands to know who he is, the first man awkwardly tried to explain he was just waiting for the bus.
Tensions escalated as the second man revealed himself as a magician and the woman as his assistant.
The woman pulled out a gun, screaming for answers.
The magician produced his own gun and began chanting from a strange relic.
The woman opens the magician's trunk and is shocked.
The first man collapsed as the spell took hold, barking and convulsing.
The woman, in anger, shoots the magician.
In response, the magician shoots the woman and they both collapse on the ground.
The man lay in melted ice cream, barking like a dog.
He crawled away on all fours, sniffing and howling.
Another woman appeared from the car, also crawling and barking.
They sniffed each other, played, and moved off together, no longer human but instead acting like dogs.